

Week 5 : Assignment 5

Detailed Solutions

Question 1

Which of the following statements is/are NOT TRUE with respect to multi-variate optimization?

- I: The gradient of a function at a point is parallel to the contours.
 - II: Gradient points in the direction of greatest increase of the function.
 - III: Negative gradient points in the direction of greatest decrease of the function.
 - IV: Hessian is a non-symmetric matrix.
- ☐ I
 - ☐ II and III
 - ☒ I and IV
 - ☐ III and IV

Solution:

The gradient of a function is **perpendicular** to the contour lines, not parallel. Therefore, statement I is false.

The gradient points in the direction of the greatest increase of the function:

$$\nabla f(x).$$

Therefore, statement II is true.

The negative gradient points in the direction of the greatest decrease:

$$-\nabla f(x).$$

Therefore, statement III is true.

For a twice differentiable function, the Hessian matrix is **symmetric**:

$$\nabla^2 f(x) = (\nabla^2 f(x))^T.$$

Therefore, statement IV is false.

Hence, the statements which are NOT TRUE are:

I and IV

Question 2

The solution to an unconstrained optimization problem is always the same as the solution to the constrained one.

- True
- ✓ False

Solution:

An unconstrained optimization problem can be written as

$$\min_x f(x),$$

with no restrictions on the value of x .

A constrained optimization problem can be written as

$$\min_x f(x) \quad \text{subject to} \quad g_i(x) \leq 0.$$

The unconstrained optimum may not satisfy the given constraints. Therefore, the solution to an unconstrained optimization problem is not always the same as the solution to a constrained optimization problem.

False

Question 3

Gradient based algorithm methods compute

- only step length at each iteration
- ✓ both direction and step length at each iteration
- only direction at each iteration
- none of the above

Solution:

Gradient-based optimization methods use the gradient $\nabla f(x)$, which provides the direction of the greatest increase of the function. For minimization, the negative gradient $-\nabla f(x)$ gives the descent direction.

A gradient-based optimization algorithm also requires a step length or learning rate. The general update rule is

$$x_{k+1} = x_k - \alpha_k \nabla f(x_k),$$

where $\nabla f(x_k)$ determines the direction, and α_k determines the step length.

Therefore, gradient-based methods compute:

Both direction and step length at each iteration

Question 4

For an unconstrained multivariate optimization problem given $f(\vec{x})$, the necessary second-order condition for $\vec{x}^*$ to be the minimizer of $f(x)$ is

- $\nabla^2 f(\vec{x}^*)$ must be negative definite.
- ✓ $\nabla^2 f(\vec{x}^*)$ must be positive definite.
- $\nabla f(\vec{x}^*) = 0$
- $f''(\vec{x}^*) > 0$

Solution:

For a stationary point $\vec{x}^*$, the first-order condition is

$$\nabla f(\vec{x}^*) = 0.$$

This alone is only necessary for a stationary point, not sufficient to classify it as a minimum.

For $\vec{x}^*$ to be a **minimizer**, the Hessian matrix $\nabla^2 f(\vec{x}^*)$ must be positive definite (all eigenvalues positive). This ensures the function curves upward in every direction around $\vec{x}^*$.

$\nabla^2 f(\vec{x}^*)$ must be positive definite

Use the following information for Questions 5, 6, 7 and 8.

$$\min_{x_1, x_2 \in \mathbb{R}} f(x_1, x_2) = x_1^2 + 4x_2^2 - 2x_1 + 8x_2.$$

Question 5

Which among the following is the stationary point for $f(x_1, x_2)$?

- $(0, 0)$
- ✓ $(1, -1)$
- $(-1, -1)$
- $(-1, 1)$

Solution:

A stationary point is obtained by setting the gradient equal to zero.

First, calculate the partial derivative with respect to x_1 :

$$\frac{\partial f}{\partial x_1} = 2x_1 - 2.$$

Now calculate the partial derivative with respect to x_2 :

$$\frac{\partial f}{\partial x_2} = 8x_2 + 8.$$

Therefore,

$$\nabla f(x_1, x_2) = \begin{bmatrix} 2x_1 - 2 \\ 8x_2 + 8 \end{bmatrix}.$$

At the stationary point, $\nabla f(x_1, x_2) = 0$. Thus,

$$2x_1 - 2 = 0 \quad \Rightarrow \quad x_1 = 1,$$

and

$$8x_2 + 8 = 0 \quad \Rightarrow \quad x_2 = -1.$$

Therefore, the stationary point is

$$\boxed{(1, -1)}.$$

Question 6

Find the eigen values corresponding to the Hessian matrix of f .

- $1, -1$
 - $1, 1$
 - ✓ $2, 8$
 - $0, 2$
-

Solution:

The Hessian matrix consists of the second-order partial derivatives. We have

$$\frac{\partial^2 f}{\partial x_1^2} = 2, \quad \frac{\partial^2 f}{\partial x_2^2} = 8,$$

and the mixed partial derivatives are

$$\frac{\partial^2 f}{\partial x_1 \partial x_2} = \frac{\partial^2 f}{\partial x_2 \partial x_1} = 0.$$

Therefore, the Hessian matrix is

$$\nabla^2 f = \begin{bmatrix} 2 & 0 \\ 0 & 8 \end{bmatrix}.$$

To calculate the eigenvalues, we solve

$$\det(\nabla^2 f - \lambda I) = 0 \quad \Rightarrow \quad \det \begin{bmatrix} 2 - \lambda & 0 \\ 0 & 8 - \lambda \end{bmatrix} = 0.$$

Thus,

$$(2 - \lambda)(8 - \lambda) = 0 \quad \Rightarrow \quad \lambda_1 = 2, \quad \lambda_2 = 8.$$

Hence, the eigenvalues are

$$\boxed{2, 8}.$$

Question 7

Find the minimum value of f .

- ☐ 0
- ☒ -5
- ☐ -1
- ☐ 1

Solution:

From Question 5, the stationary point is $(x_1, x_2) = (1, -1)$. Substitute these values into

$$f(x_1, x_2) = x_1^2 + 4x_2^2 - 2x_1 + 8x_2.$$

Therefore,

$$f(1, -1) = (1)^2 + 4(-1)^2 - 2(1) + 8(-1) = 1 + 4 - 2 - 8 = -5.$$

Also, from Question 6, the Hessian eigenvalues are $\lambda_1 = 2 > 0$ and $\lambda_2 = 8 > 0$. Since both eigenvalues are positive, the Hessian is positive definite, so the stationary point is a minimum.

Hence, the minimum value is

$$\boxed{-5}.$$

Question 8

What is the minimum value of $f(x_1, x_2)$ subject to the constraint $x_1 + 2x_2 = 7$?

- -5
- -1
- ✓ 27
- 0

Solution:

We are given

$$f(x_1, x_2) = x_1^2 + 4x_2^2 - 2x_1 + 8x_2 \quad \text{subject to} \quad x_1 + 2x_2 = 7.$$

From the constraint, $x_1 = 7 - 2x_2$. Substitute this into the objective function:

$$f(x_2) = (7 - 2x_2)^2 + 4x_2^2 - 2(7 - 2x_2) + 8x_2.$$

Expanding, $(7 - 2x_2)^2 = 49 - 28x_2 + 4x_2^2$, so

$$f(x_2) = 49 - 28x_2 + 4x_2^2 + 4x_2^2 - 14 + 4x_2 + 8x_2.$$

Combining like terms,

$$f(x_2) = 8x_2^2 - 16x_2 + 35.$$

Differentiate with respect to x_2 :

$$\frac{df}{dx_2} = 16x_2 - 16.$$

For a stationary point, $16x_2 - 16 = 0$, so $x_2 = 1$.

Using the constraint, $x_1 + 2(1) = 7$, so $x_1 = 5$. Therefore, $(x_1, x_2) = (5, 1)$.

Now substitute into the objective function:

$$f(5, 1) = (5)^2 + 4(1)^2 - 2(5) + 8(1) = 25 + 4 - 10 + 8 = 27.$$

Therefore, the minimum value is

$$\boxed{27}.$$

Question 9

Find the maximum value of $f(x, y) = 49 - x^2 - y^2$ subject to the constraint $x + 3y = 10$.

- 49

- 46
- 59
- ✓ 39

Solution:

From the constraint $x + 3y = 10$, we obtain $x = 10 - 3y$. Substitute this into the objective function:

$$f(y) = 49 - (10 - 3y)^2 - y^2.$$

Expanding, $(10 - 3y)^2 = 100 - 60y + 9y^2$, so

$$f(y) = 49 - (100 - 60y + 9y^2) - y^2 = 49 - 100 + 60y - 9y^2 - y^2.$$

Therefore,

$$f(y) = -51 + 60y - 10y^2.$$

Differentiate: $\frac{df}{dy} = 60 - 20y$. Setting the derivative equal to zero,

$$60 - 20y = 0 \quad \Rightarrow \quad y = 3.$$

Using the constraint, $x + 3(3) = 10$, so $x = 1$. Therefore, $(x, y) = (1, 3)$.

Now calculate the value of the objective function:

$$f(1, 3) = 49 - (1)^2 - (3)^2 = 49 - 1 - 9 = 39.$$

Therefore, the maximum value is

$$\boxed{39}.$$

Question 10

Consider the optimization problem

$$\min_{x,y} x^2 - xy + y^2$$

subject to the constraints

$$2x + y \leq 1, \quad x + 2y \geq 2, \quad x \geq -1.$$

Find the Lagrangian function for the above optimization problem.

- $L(x, y, \mu_1, \mu_2, \mu_3) = x^2 - xy + y^2 + \mu_1(2x + y - 1) + \mu_2(x + 2y - 2) + \mu_3(-x - 1)$
- ✓ $L(x, y, \mu_1, \mu_2, \mu_3) = x^2 - xy + y^2 + \mu_1(2x + y - 1) + \mu_2(2 - x - 2y) + \mu_3(-x - 1)$

- $L(x, y, \mu_1, \mu_2, \mu_3) = x^2 - xy + y^2 + \mu_1(2x + y - 1) + \mu_2(x + 2y - 2) + \mu_3(x + 1)$
- $L(x, y, \mu_1, \mu_2, \mu_3) = x^2 - xy + y^2 + \mu_1(1 - 2x - y) + \mu_2(2 - x - 2y) + \mu_3(-x - 1)$

Solution:

The objective function is

$$f(x, y) = x^2 - xy + y^2.$$

For constructing the Lagrangian, we write all inequality constraints in the standard form $g_i(x, y) \leq 0$.

First constraint: $2x + y \leq 1$. Therefore,

$$2x + y - 1 \leq 0 \quad \Rightarrow \quad g_1(x, y) = 2x + y - 1.$$

Second constraint: $x + 2y \geq 2$. Multiplying by -1 ,

$$-x - 2y \leq -2 \quad \Rightarrow \quad 2 - x - 2y \leq 0 \quad \Rightarrow \quad g_2(x, y) = 2 - x - 2y.$$

Third constraint: $x \geq -1$. This can be written as

$$-x \leq 1 \quad \Rightarrow \quad -x - 1 \leq 0 \quad \Rightarrow \quad g_3(x, y) = -x - 1.$$

The general Lagrangian is

$$L(x, y, \mu_1, \mu_2, \mu_3) = f(x, y) + \mu_1 g_1(x, y) + \mu_2 g_2(x, y) + \mu_3 g_3(x, y).$$

Substituting the objective function and constraints,

$$L(x, y, \mu_1, \mu_2, \mu_3) = x^2 - xy + y^2 + \mu_1(2x + y - 1) + \mu_2(2 - x - 2y) + \mu_3(-x - 1)$$

Answer Summary

Q	Correct Answer
1	I and IV
2	False
3	Both direction and step length at each iteration
4	$\nabla^2 f(\vec{x}^*)$ must be positive definite
5	$(1, -1)$
6	2, 8
7	-5
8	27
9	39
10	$L = x^2 - xy + y^2 + \mu_1(2x + y - 1) + \mu_2(2 - x - 2y) + \mu_3(-x - 1)$